GridPing: Bridging Grid Edge Technologies and Smart Meter Data Through Accessible Power Quality Visualization

LESLY ROJAS CALOCA*, DANIEL NIKOLOV†, AND EDUARDO COTILLA-SANCHEZ.‡

[1]School of Electrical Engineering and Computer Science, Oregon State University, Corvallis, OR 97331 USA

CORRESPONDING AUTHORS: Lesly Rojas Caloca & Daniel Nikolov (E-mail: gridpingapp@gmail.com)

Source code and demonstration are available at: https://gridping.vercel.app/

ABSTRACT This research paper presents the design concept of GridPing, a lightweight smart meter app that translates complex grid data into intuitive insights utility clients can access on their smartphone. Through this use scenario, utility customers are empowered to make decisions based on the grid state, such as disconnecting an electric vehicle charger when power quality is poor, thus, decreasing electricity costs. This research leverages the functionality and architecture of smart meters within modern power systems to emphasize the smart meter's role in monitoring power quality, and supports computation for distributed energy resources. Importantly, these end-user actions would aid utilities by alleviating strain on the grid during periods of congestion. Our work emphasizes the importance of power quality, particularly for sensitive systems such as medical equipment and electric vehicles, where voltage fluctuations, harmonics, or outages can impact performance and reliability. Special attention is given to scenarios such as outages and islanding, where timely information can help prevent grid complications for everyday users and high-stakes environments. GridPing demonstrates how bridging smart meter data with accessible application design can improve energy awareness and support critical infrastructure by enabling more informed and responsive energy decisions.

INDEX TERMS Grid Edge, Power Grid, Power Quality, Smart Meter, Software

I. MOTIVATION & SIGNIFICANCE

THE rapid growth of infrastructure electrification in residential areas has introduced new challenges for utilities. Notably, these include maintaining reliable high-quality power and communicating the grid status to customers. Shifting from the provider's perspective to the consumer's, for many people, devices such as electric vehicles (EVs) and at home medical equipment (such as infusion pumps, oxygen concentrators, and continuous positive airway pressure machines) are essential. These technologies rely on having stable *high-quality power*.

Peak demand from electric vehicles, combined with increasing household loads, can introduce fluctuations in voltage, frequency, and waveform quality [1]. These issues are often invisible to users; however, they can reduce device performance, shorten equipment lifespan, or, in some cases, create safety risks without ever alerting the user [2].

Most currently installed smart meter systems are designed to track how much power is used, not the *quality of power* being delivered. This work is motivated by the need to make power quality visible and understandable at the user level.

A partner electrical utility also recognized this issue. Throughout 2025, our research team held discussions with this utility and presented the findings of a related island detection research project. During these meetings, we had conversations with experts in the domain regarding their push for smart meters with greater computational abilities. Their plan included leveraging this computation to create connected applications to improve end user interactions at the grid edge. With this knowledge in mind, the research pivoted from a pure analysis of islanding algorithms to the production of a user-facing application that conducts grid analysis at the meter level. This later led to the development of GridPing, a smartphone application which provides users with live and historical power quality telemetry through their location and the grid as a whole. The application receives its data directly from the smart meter and notifies users when it is cost effective to perform strenuous grid activities, such as charging EVs or using home appliances. In addition, it provides historical outage and power quality data, allowing users to see changes over time. However, this also benefits utilities by ensuring that users avoid strenuous activities when the grid is under strain and power is expensive, improving grid resilience.

The significance of this work is based on the foundation of people and infrastructure. A main focus of this research is to increase the accessibility of power grid information to the benefit of customers and the utility. A main component of this push is including access to cost knowledge with the goal of empowering users to make informed decisions regarding their electricity usage. This comes as a response to how "The main complaint of domestic customers concerned the costs which were perceived too high, especially where cross-subsidizing was used to keep prices low for industrial or agricultural customers." [3]. Improving access to power quality information has the potential to increase reliability, protect critical devices, and support safer and resilient communities as electrification continues to grow.

II. DEFINITIONS

A. Power Quality

Power quality typically refers to maintaining the near sinusoidal waveform of power distribution, bus voltages, and currents at rated magnitude and frequency (often used to express voltage quality, current quality, reliability of service, and quality of power supply) [4]. While there are many metrics which can be used to represent power quality, a simple yet still accurate one was selected for this application. Difference in per unit voltage magnitude (p.u.) from the expected grid state was the metric used to represent power quality for the backend implementation. Section 3.C, details how this value is transformed into a percentage which accessible for non-technical users.

In a perfect system, power is steady, but that is not always the case in a real-world scenario. Issues like voltage drops, spikes, outages, grid islanding, and outages affect the power quality that individuals encounter. These problems matter more today because we rely on sensitive technology. For medical devices, it is critical for these systems to be stably powered to operate safely, as small disturbances can affect their performance. Power quality is also critical in electric vehicles since they can be a large load on the actual power grid. During peak hours, when electricity demand is highest, EV chargers place more stress on the grid, increasing the risk of power quality issues. Maintaining high power quality is essential to ensure reliable operation, efficient energy use, and safe performance of modern systems.

B. Smart Meters

Like traditional meters, smart meters measure electricity usage in real time, however, they provide more functionality, communication capabilities, and detailed data than traditional meters. In addition to tracking energy consumption, they can monitor grid conditions such as voltage and power quality. Smart meters can also leverage communication channels between the user and the utility company, however, after sampling electrical utility practices, *most* of the data is not currently shared with customers. Bridging this gap with GridPing would create awareness of energy and grid status to costumers, especially with the increasing demand of electric vehicles and overall grid access.

III. GridPing: BACKEND IMPLEMENTATION

A. General Overview

The greater computational ability of next-generation smart meters enables the GridPing backend to execute real-time data processing. However, it is still imperative that the GridPing application remain as lightweight and efficient as possible, since it would not be the only application running on the updated meters. Memory is one of the major constraints established by vendors. Unfortunately, most industry standard grid analysis software are quite computationally demanding. Although MATPOWER and PowerWorld are generally considered lightweight programs in transmission environments [5], for our purposes they are quite computationally demanding or require additional browser infrastructure. Running these programs on a browser would introduce additional cybersecurity implications and result in a simplified version that does not take full advantage of the existing libraries and ecosystem. For this reason, the lightweight Julia programming language and the accompanying Powermodels.jl library were selected to perform the power grid analysis.

Since Powermodels.jl is able to import (parse) MATPOWER case files, we continued using the well known Poland2383wp.m test case file. This Poland case file was treated as an ideal grid state, which was algorithmically modified to generate 24 hourly case files and manually modified to create several grid event cases. These cases are used to imitate the smart meter's reception of global grid states from the utility and, by extension, the local meter reading at the user's bus. The whole case file is used in calculating the global state, whereas exclusively voltage (p.u.) through the smart meter running the code is used to calculate the local state. Backend users can decide how frequently case files, treated as simulated meter readings, are received and processed.

In this illustrative example, upon receiving a case file, the data processing algorithm compares the current state to the ideal grid state by simulating the feasibility of the power grid, running the island detection algorithm, calculating the power quality, and ultimately publishing the results to a local and/or remote database. Figure 1 illustrates this data flow along with how the frontend application receives database records through AWS Lambda function calls.

The proposed communication layer, implemented through the AWS PostgreSQL database, is essential for the design constraints of GridPing. It enables asynchronous communication between the smart meter, user devices, and the utility, while also allowing the meter's locally hosted database to be optional, freeing more space on the meter.

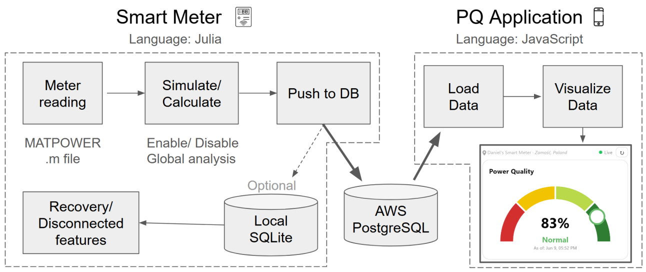

FIGURE 1. **Flow of data from the backend to the frontend, through the AWS hosted PostgreSQL database**

B. Generation of Meter Readings

The 24 hourly case files were generated by altering the voltages through each node according to a randomized general demand distribution. When processed sequentially, these cases mimic how voltages change throughout the day, with less demand throughout the night and greater demand as users wake up and when they return home.

In addition to selecting how often a new case file is processed, a backend user can also determine the probability of an event case occurring at every meter reading. Event cases represent an altered grid state in which a large power quality impacting event has occurred. These events range from a small and/or large island being invoked, to a cascading global outage, to failure at the Demo's examined bus number 6. These files were created through manual deletion and modification of buses and branches in the Poland2383wp.m case file. Their inclusion allows for the GridPing application to demonstrate how it handles a variety of off-nominal states.

C. Power Quality Metric Calculations

- **Local Metric** As discussed prior, we use a simple metric as a proxy to power quality. This value is calculated by subtracting the voltage through the smart meter from the corresponding bus' voltage in the Poland2383wp.m case file. A value of 0 indicates no change from the ideal grid state, whereas positive and negative values indicate greater or lower voltage values respectively. This value, upon reception by the front-end, is converted to a 0% to 100% health score derived from the bus voltage deviation in per-unit. 0.00 p.u. maps to 100% and deviations beyond -0.20 p.u. (undervoltage) or $+0.10$ p.u. (overvoltage) map to 0%, with an asymmetric penalty curve following ANSI C84.1 tolerance bands [6]. It is important to note that global power quality is calculated by examining the mean power quality through each node.
- **Global Metrics** Another power quality adjacent metric is convergence. This indicates whether the optimal power-flow solver (`runopf()` function) from PowerModels.jl was able to reach a physically possible grid state. These simulations take place on the smart meter side of the GridPing application, where a value of True is returned for converged states, whereas False indicates a non-converged state. Non-convergence is a strong indicator for grid risk, and thus a useful metric, traditionally in the control room, to translate for users. Islanding occurs when a section of the power grid becomes disconnected from the main grid, resulting in multiple independent connected components. Upon completion of the grid simulation, the current grid structure is checked for islands using the `calc_connected_components()` function of the PowerModels.jl library. Values greater than one are abnormal for the Poland case, however, the backend is easily configured to accommodate other grid standards depending on how many islands are expected. Although not always immediately problematic, unintentionally created islands can place certain sections of the grid under unexpectedly high strain and lead to cascading outages.

The finer details of how these values are calculated can be seen in the meter_reading_simulator.jl file (found in the GridPing-Backend Github repository). This is the main file which handles the flow of case files and the publishing of records to databases.

D. Database Structure

Once power quality metrics are obtained, they must be stored as records in a database to enable access by the frontend. At the end of the meter-side data processing flow, two SQL INSERT queries are constructed, one record for the smart meter's local data and another for the grid-wide data. The database schema, visible in Figure 2, demonstrates how these values are stored. In this architecture, buses are treated as individual meters.

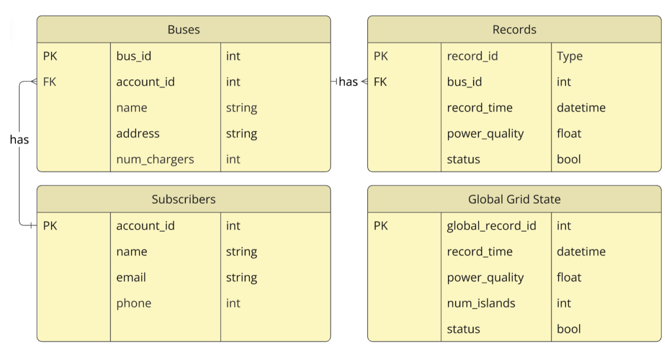

FIGURE 2. **GridPing Database Schema**

For simplicity, this structure has limited consideration of other smart meters calculating the global state in parallel. However, time-based lookup queries easily be implemented to prevent duplication of records and compute, however, this assumes that all meters receive the same global case file.

E. Database Hosting

When designing the data pipeline, a meter-hosted SQLite database was chosen due to lightweight nature of SQLite. It was decided that the required communication between the smart meter and frontend device could become a security risk, limit database storage, and require additional communication infrastructure. A cloud-hosted database resolved these issues while being independent of the state of the smart meter. AWS RDS was selected as the database hosting platform for GridPing due to its use by the consulted utility. The service hosts a PostgreSQL database, allowing for the local SQLite database to be optional, however, enabling it allows for utilities to retain a local store of the last 24 hours of records in case of an inability to reach cloud services. This feature allows the application to continue operating in the event of a service outage and sync the databases once connection is reestablished. In the code, these options appear as two flags, one for SQLite and another for AWS PostgreSQL.

F. Serving Data to the Frontend

For security purposes, an AWS Lambda request handler API was implemented between the user application and the database. Once a record is successfully stored on AWS, the data are packaged in JSON format, and the response is served to the frontend upon HTTP request. This standard practice prevents the exposure of credentials and prevents arbitrary queries from being run against the database. A copy of the function is also maintained in the GridPing-Backend Github repository with further details.

G. Running the Backend

Simulation of the smart meter data flow is accomplished by connecting the desired device to AWS through the terminal. Then, the meter_reading_simulator.jl is configured and executed start populating the database, allowing the GridPing end-user application to display new grid states. At the time of publishing, a permanently running instance of the backend code is being implemented on the NVIDIA Jetson Orin Nano developer kit. This device, with similar hardware to next-gen meters, will enable persistent data flow on the GridPing website.

IV. GridPing: FRONTEND IMPLEMENTATION

A. General Overview

The objective of this app is for power quality information to be accessible to customers. In order to accommodate different demographics and expand how the data is presented, a two-mode structure was proposed: "Simple" and "Advanced" Mode.

The main visual feature for simple mode is a dial showcasing the power quality at the user's location by using a gradient from red to green. The main visual features for advanced mode are graphs and tables displaying more in depth power quality data.

B. Simple Mode

Simple mode is intended for use by all users, allowing them to receive a quick summary of their household's current power quality and grid forecast. If a user owns an electric vehicle, they will also see a similar summary informing them of the most cost effective time to charge their vehicle, alongside notifications of when to plug in or unplug their vehicle.

We used React's useState and useEffect hooks to manage application state and synchronize the user interface with the backend implementation. When new power quality measurements are retrieved, the state is updated, triggering a re-render of the relevant components and ensuring that the displayed information remains current.

- **Power Quality Dial** A dial component as shown in Figure 3 displays the current power quality level using color-coded ranges, allowing clients to quickly assess grid conditions without interpreting numerical data. As described prior, the power quality percentage is a 0% to 100% health score derived from the bus voltage deviation in per-unit. Additional components, such as outage notification, appliance information, and charging recommendations, use the incoming meter data to generate actionable insights rather than technical metrics. The dial is sectioned as following:
  - 0% 'Critical'
  - 1-19% 'Brownout Risk'
  - 20-49% 'Poor'
  - 50-79% 'Fair'
  - 80-94% 'Normal'
  - 95-100% 'Ideal'
- **EV Charge Forecast Card** This card translates power quality and voltage conditions into charging recommendations for electric vehicle owners. Based on current grid conditions, the card categorizes charging suitability into five states:
  - 0–19% Do Not Charge | Critical High/Low Voltage
  - 20–49% Avoid Charging | Well Above/Below Normal Voltage
  - 50–79% Charge if Necessary | Above/Below Normal Voltage
  - 80–94% Good Time to Charge | Slightly Low/High Voltage
  - 95–100% Best Time to Charge | Ideal Conditions

  Each state is paired with a color-coded indicator reflecting the severity of the grid conditions. Allowing clients to quickly determine whether current conditions are suitable for EV charging without needing to interpret raw data and notifying them of the most cost effective time to charge (during off-peak hours).
- **Regional Power Quality** Takes into consideration the state of the client's entire distribution grid. Using the following display states:

- 0–19% Critical | Significantly higher/lower voltage than expected
- 20–49% Poor | Well above/below normal voltage
- 50–79% Fair | Slightly higher/lower voltage than expected
- 80–94% Normal | Near-ideal voltage conditions
- 95–100% Ideal | Ideal voltage conditions

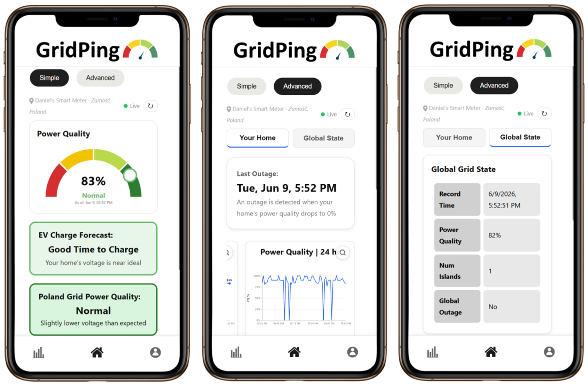

FIGURE 3. Simple Mode (Left) and Advanced Mode (Middle & Right)

C. Advanced Mode

Unlike the simplified dashboard presented in Simple Mode, Advanced Mode provides technical metrics, historical trends, and network-level information that supports deeper analysis of the client's data.

This interface includes interactive visualizations of the *last 3 meter readings, power quality over 24 hours, last outage occurrence* and the *global state grid*, allowing clients to monitor changes over time, identify potential abnormalities, and be informed of their power usage.

Historical data is displayed through line graphs and tables that present the most recent meter readings, along with time stamps that cover both the clients home status and the state of the global grid.

GridPing uses React and Vite to create the interface. The application is divided into reusable components, such as graphs, data cards, and navigation elements. Opening the application for future updates and added features.

SVG-based graphs are placed in a horizontal carousel with CSS scroll snapping were used to maximize the limited screen space while still providing access to multiple visualizations. The graph scales and data points are generated dynamically based on the incoming dataset, allowing the visualizations to adapt to changing conditions.

V. FUTURE APPLICATIONS OF WORK

A. GridPing Dashboard for Grid Control Rooms

Although the main focus of this research was on improving the experience of users at the grid edge, the GridPing infrastructure was designed with a broad range of potential use cases in mind. Figure 4 demonstrates this through a frontend only prototype intended for use in power grid control rooms. If this concept were continued into another fullstack application, much of the data flow would remain unchanged, however, more metrics would be added to better match the needs of control room engineers. These include relevant weather factors, Voltage vs. Bus graphs, and highlighting regions at risk of becoming islanded. The purpose of this prototype was to demonstrate how the GridPing backend could be used as a base to develop a variety of power grid applications.

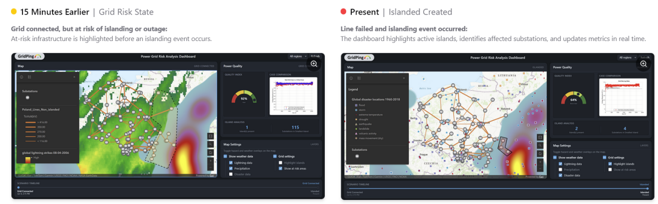

FIGURE 4. GridPing Control Room Application States

VI. CONCLUSION

GridPing demonstrates how modern metering infrastructure and software can be used to improve the accessibility of smart meter data for clients. By combining a React based frontend with cloud hosted data services, the application provides active insights into power quality, voltage conditions, and outage events through a responsive interface. The implementation of data visualizations, user modes, and dashboard components allows clients with varying levels of technical knowledge to better understand the status of their electrical systems. Throughout the development process, emphasis was placed on creating a lightweight, secure, scalable, and maintainable architecture that supports future expansion. As metering infrastructure continues to expand, platforms capable of processing and visualizing large-scale metering data in real time will play an important role in improving grid resilience, operational efficiency, and customer awareness.

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